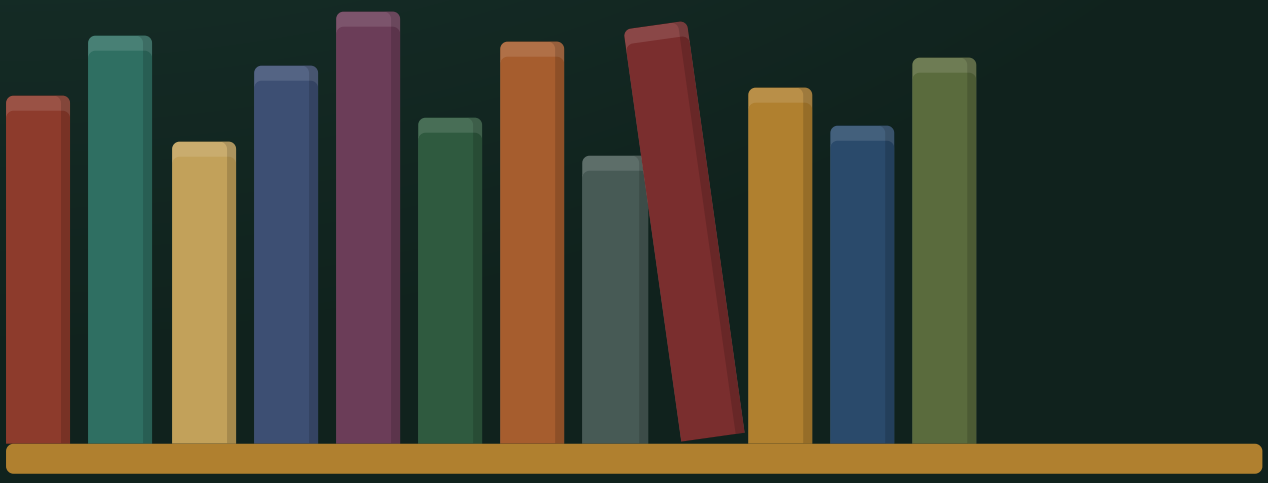
USER MANUAL

Library Management

Find, borrow and manage books – for readers, librarians and administrators.



Program version 3.0 · As of October 2026 Project Catalytix · developed by helfi.space · free software (GNU AGPL v3)

Contents

1 **Overview** 3

The three roles 3
The layout of the interface 3
Language, help and what's new 4

2 **Signing in** 6

With username and password 6
The second factor 6
On a smartphone 7
If you cannot sign in 8

3 **For everyone: finding and borrowing** 9

The overview 9
Searching the catalogue 9
Borrowing a book 10
Reservations 11
My loans 12
Profile and emails 13
Setting up two-factor authentication 14
Registering yourself 15

4 **For librarians: collection and lending** 16

Adding a book 16
Quick capture 17
Locations 17
Editing and deleting books 18
Lending from the catalogue 18
The circulation desk 19
Managing reservations 21
All loans and overdue notices 21
Group librarians 22

5 **For administrators** 23

Users 23
Solving sign-in problems 24
Groups and reference-only stock 25
Active Directory 26
Email 27
Branding 28
Settings 29
Two-factor and internal network 30
Retention and data protection 31
Backup 32

6 **Operations for IT** 34

Installation 34
Updating 34
Fixing a version and going back 35
Moving from versions before 3.0 35
Restoring 35
Emergency commands 36

7 **Frequently asked questions** 37

CHAPTER 1

Overview

The library management system runs in your browser – on a PC, a laptop or a smartphone. There is nothing to install. This manual explains all functions, sorted by what you are allowed to do in the program.

The three roles

Every account has exactly one role. It determines which menu items you see.

Reader

Search for books, borrow and renew them yourself, and keep track of your own loans.

Chapters 2 and 3

Librarian

Add and maintain books, lend and take back books at the circulation desk, follow up on overdue loans.

Chapters 2 to 4

Administrator

Everything – plus users, groups, sign-in via Active Directory, email, security and backup.

All chapters

Groups link books and people, for example *IT apprenticeships* or *Office management*. A book either belongs to one group or is **general**. Anyone can borrow general books; group books can only be borrowed by members of that group.

The layout of the interface

On the left is the **sidebar** with all areas, at the top the **search bar** for the catalogue. At the very bottom of the sidebar you will find your name, the switch for light and dark mode, and **Sign out**. Numbers in red circles point to something that needs attention – such as overdue loans or new registrations.

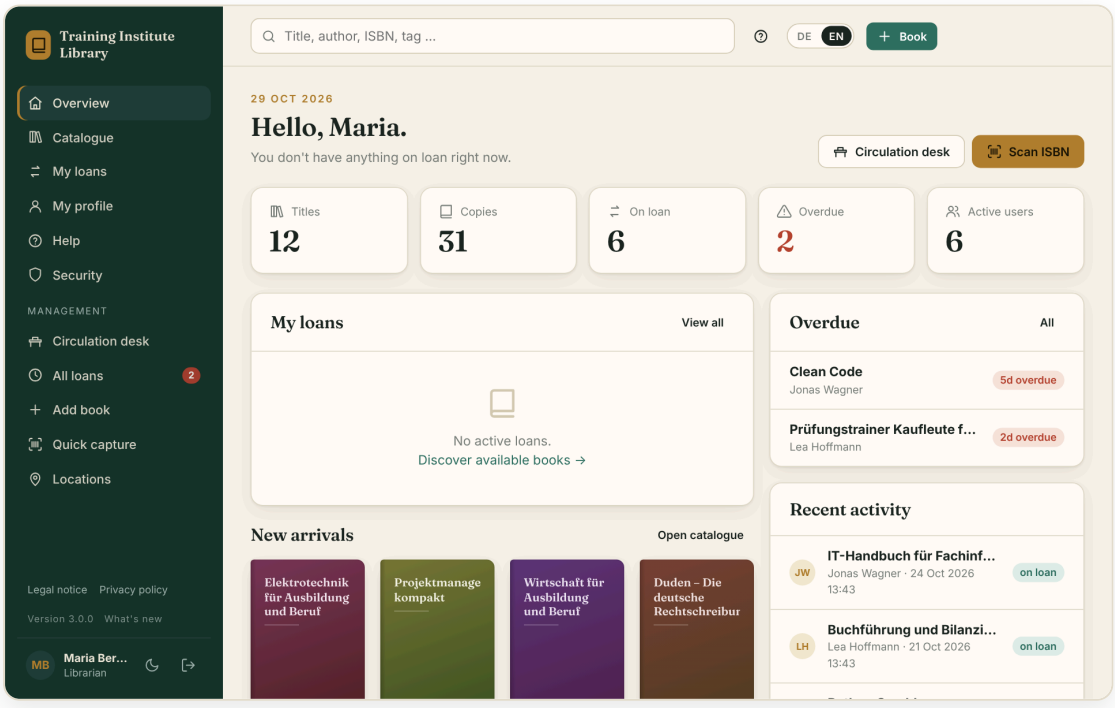


Fig. 1 – The overview as seen by a librarian: sidebar on the left, key figures and overdue loans at a glance.

> **TIP**
> Below the sidebar menu are the links to the legal notice and privacy policy, along with the program version. Clicking the version shows the licence and source code information.

Language, help and what's new

The program is available in **German and English**. Switch at the top right with DE / EN – even on the sign-in page. Once you are signed in, your choice is saved in your profile and then applies on every device and to all emails sent to you. If you don't choose, you get your browser's language; if neither matches, the default language set by the administrator under *Settings → General* applies.

Open the **help** via Help in the sidebar or the **?** at the top right. The question mark jumps straight to the section for the page you are on. The help only shows what belongs to your role and follows your library's settings – if it says “Loan period 28 days”, that really is your loan period. Use the search box to find the right topic quickly; this manual is available there as a PDF download.

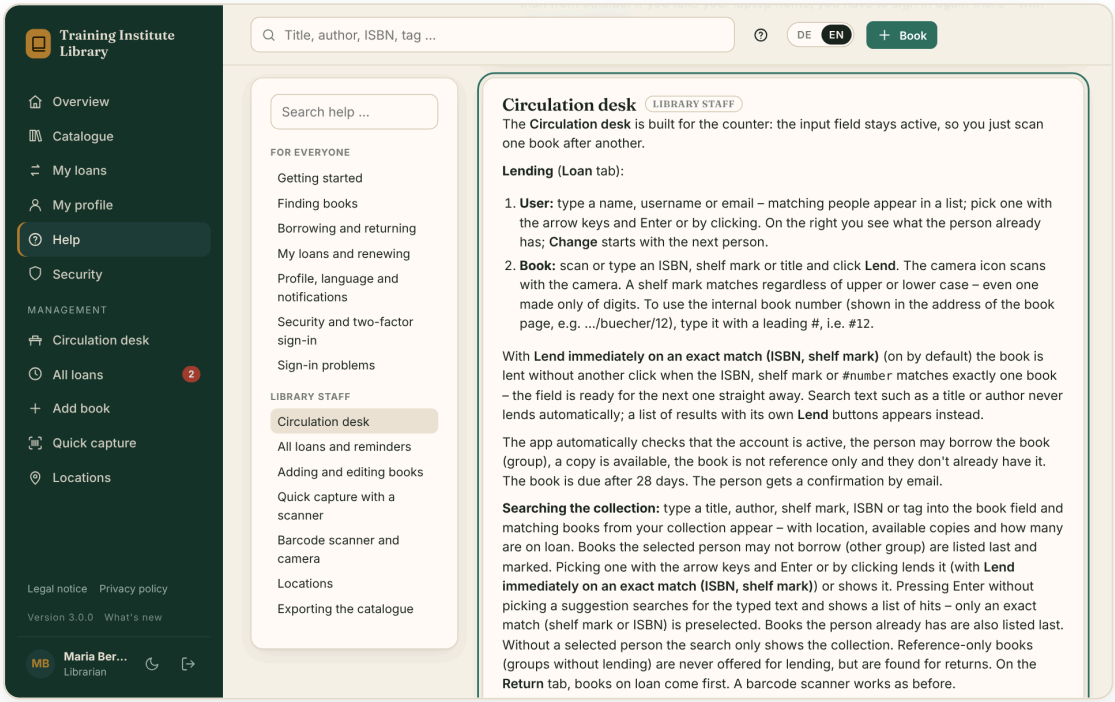


Fig. 2 – The help as seen by a librarian: the topics for her role on the left, the section on the circulation desk on the right.

What's new (at the bottom of the sidebar next to the version number, and in the user menu) explains briefly what changed in recent versions – new features, improvements and security updates. Everyone only sees what concerns their role. After an update, a small dot and a slim line on the overview point to news; once viewed or dismissed, they disappear until the next update.

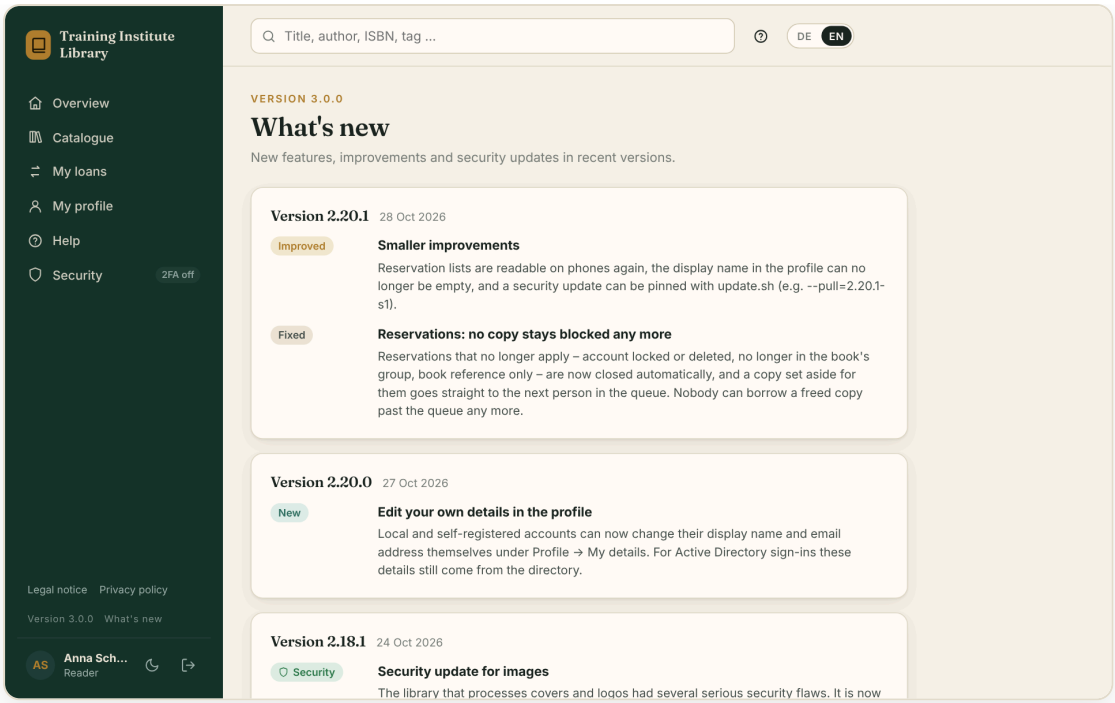


Fig. 3 – What's new, as seen by a reader.

CHAPTER 2

Signing in

The library can be reached on the internal network and over the internet. Depending on where you connect from and which role you have, a second factor is also requested.

With username and password

1. Open the library's address in your browser (the library will give you the exact address).
2. Enter your **username** and **password**. If you have an account on the institute's network, these are the same credentials you use to sign in to your PC.
3. Tick **Stay signed in** if you like – then you won't have to sign in again on this device for 14 days. Please don't tick it on other people's or shared devices.
4. Click **Sign in**.

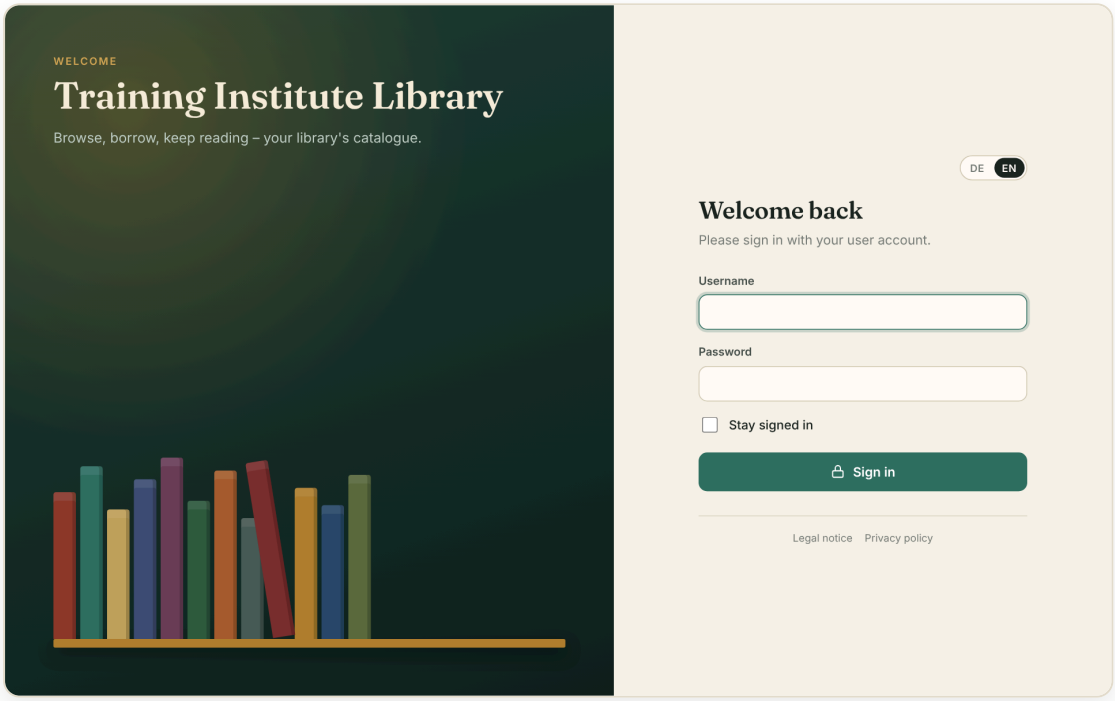


Fig. 4 – The sign-in page.

The second factor

If you have set up an authenticator app or a security key (see section 3.7), a second step follows after the password:

- **Authenticator app:** open the app on your phone and type in the current 6-digit code.
- **Security key (e.g. YubiKey):** click **Confirm with security key**, insert the key and touch it when it flashes.
- **No access?** With **Use a recovery code** you sign in with one of your emergency codes.

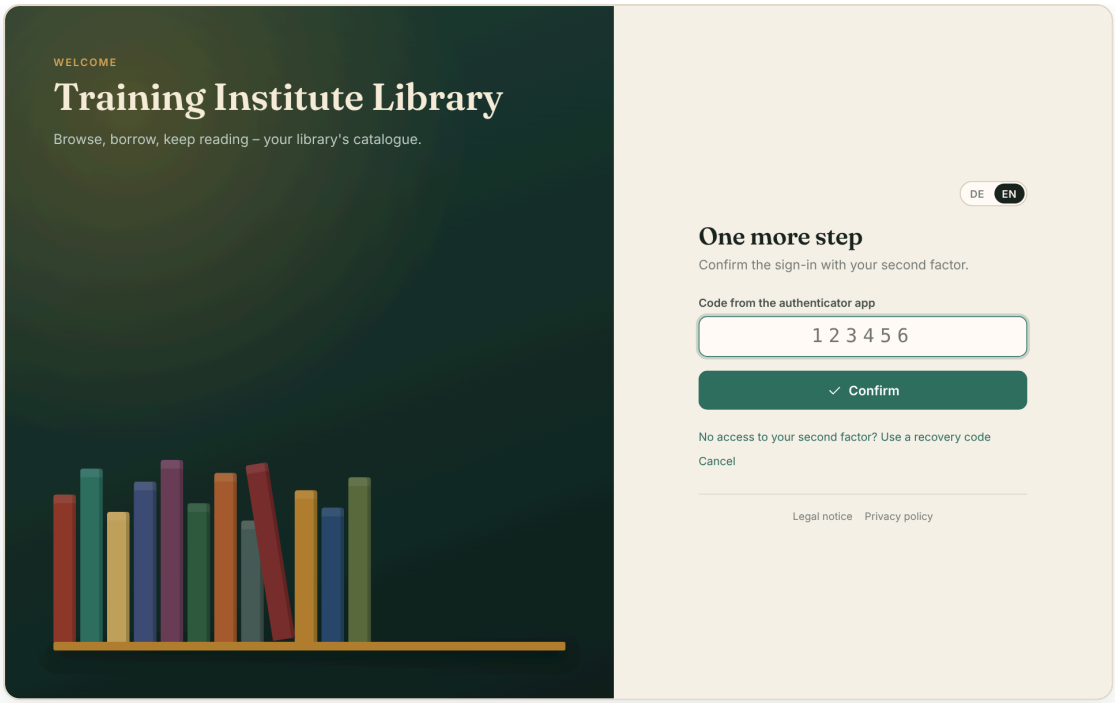


Fig. 5 – Second sign-in step with a code from the authenticator app.

> **GOOD TO KNOW**
>
> Depending on the settings, the second factor may only be requested for access **from outside** (internet), but not on the internal network. If you sign in on site without a second factor and then take your laptop home, you have to sign in again there – this time with your second factor.

On a smartphone

On a phone the page adapts automatically. Open the sidebar with the menu icon ☰ at the top left; your profile, light/dark mode and sign-out are in the round icon with your initials at the top right.

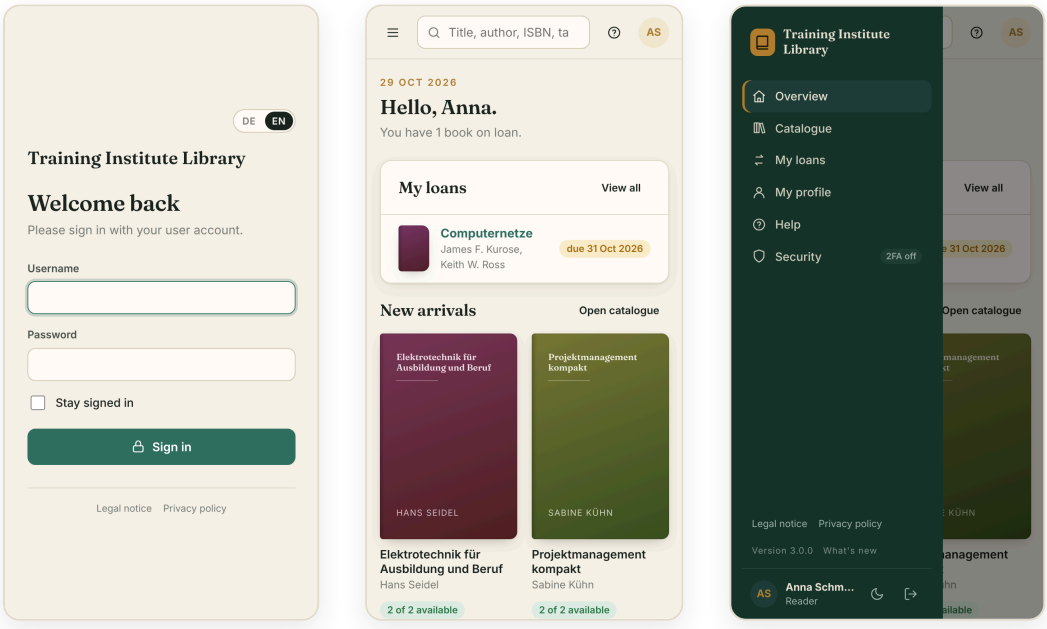


Fig. 6 – Sign-in, overview and open menu on a smartphone.

If you cannot sign in

MESSAGE / SITUATION	WHAT TO DO
“Incorrect username or password.”	Check your spelling (Caps Lock?). For security reasons, the page does not say whether the account exists or whether it is locked.
“Too many failed attempts”	After 8 failed attempts, sign-in is locked for 5 minutes. Wait – or an administrator can lift the lock straight away.
Forgotten password	For institute network accounts, IT resets the password. For local accounts, an administrator sets a new password. If you still know your password, you can change it yourself under *My profile*. There is deliberately no “forgot password” function.
Lost the phone with the authenticator app	Sign in with a recovery code or ask an administrator to reset your two-factor authentication.
“The authenticator app for this account needs to be set up again”	Sign in with a recovery code or security key and set up the app again under *Security*; delete the old entry in the app.

CHAPTER 3

For everyone: finding and borrowing

This chapter applies to all roles. Readers only see the areas described here.

The overview

After signing in you land on the **Overview**. It shows your current loans with their due dates and the books most recently added. The colour of the due-date label tells you the status: **green** = still time, **yellow** = due soon, **red** = overdue.

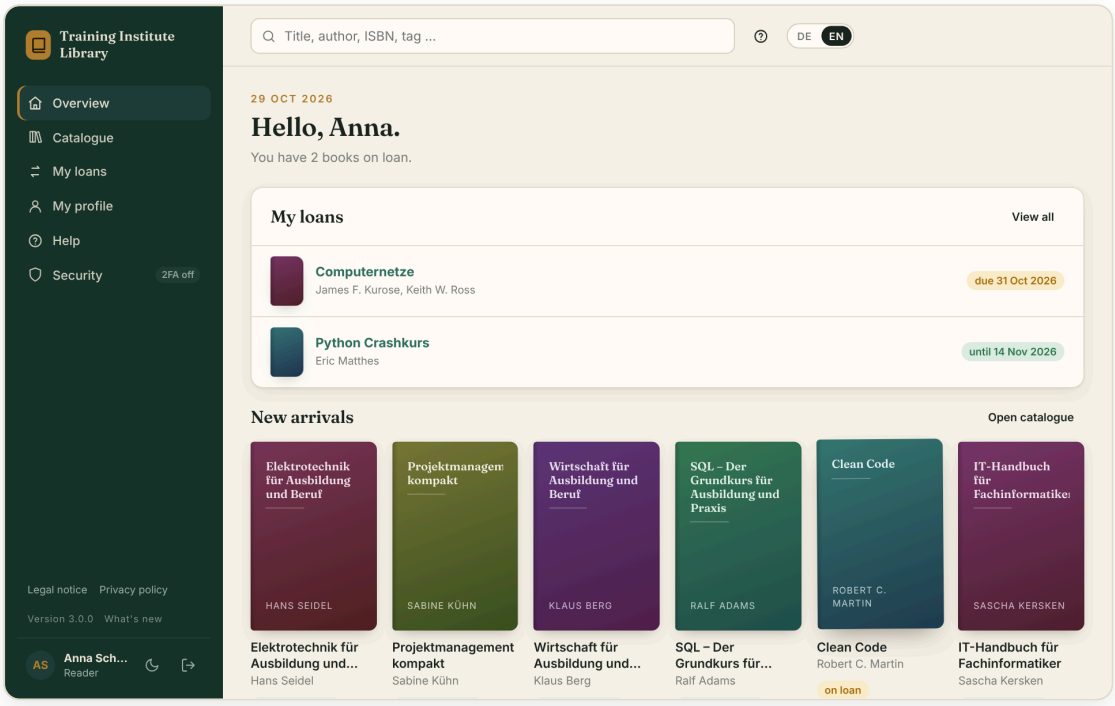


Fig. 7 – A reader's overview.

Searching the catalogue

Under **Catalogue** you will find all the books you are allowed to see. The search box at the top searches title, subtitle, author, publisher, tags, category, shelf mark, location and ISBN – you can also type or scan an ISBN directly. Matching books with author, shelf mark and available copies appear while you type (from two characters); a click or the arrow keys and Enter open the book directly, Enter without a choice shows all hits.

- **Filters:** by group, location and status (*available* / *on loan*). Books that are **reference only** are marked – they are for use on site and are not lent out.
- **Sorting:** title, author, recently added or year published.
- **View:** switch between tiles with book covers and a compact list. Your browser remembers the choice.

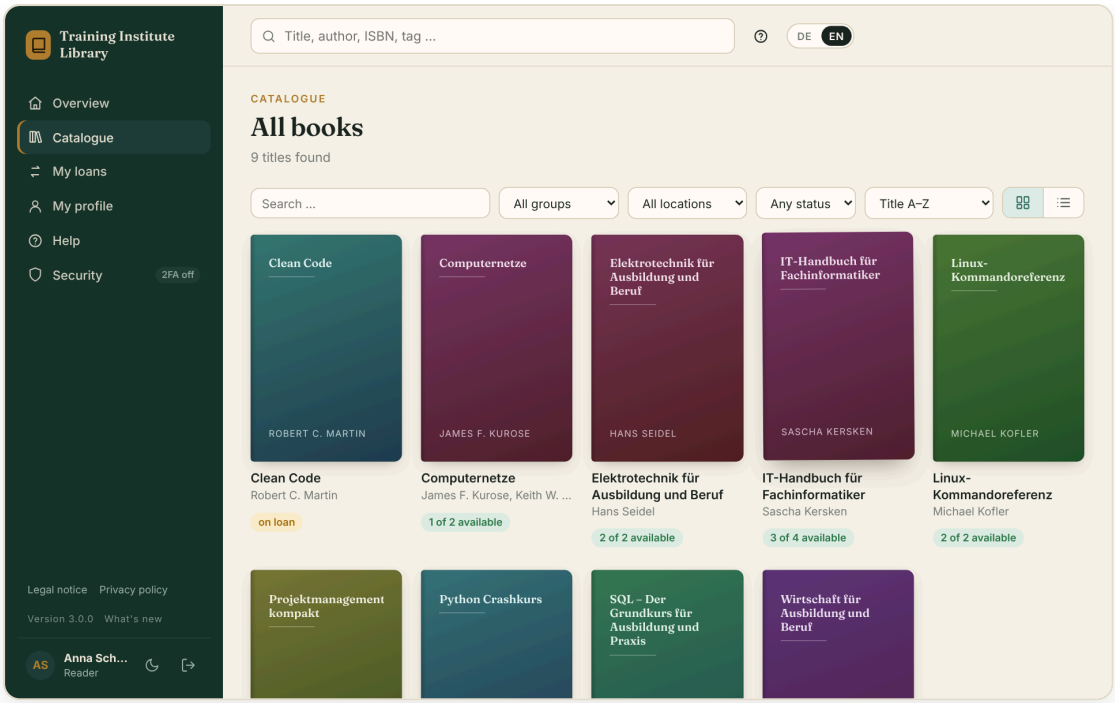


Fig. 8 – The catalogue in tile view with filters.

> **NOTE**
> Whether you see all books or only those of your groups plus the general ones is decided by the library. Either way, you can only borrow books of your groups and general books.

Borrowing a book

1. Open the book by clicking its cover or title. The book page shows all details, its location on the shelf and how many copies are available.
2. Click **Borrow for myself**. The loan period is shown just above it (default: 28 days).
3. Take the book from the shelf at the location shown. If email is set up, you will receive a confirmation.

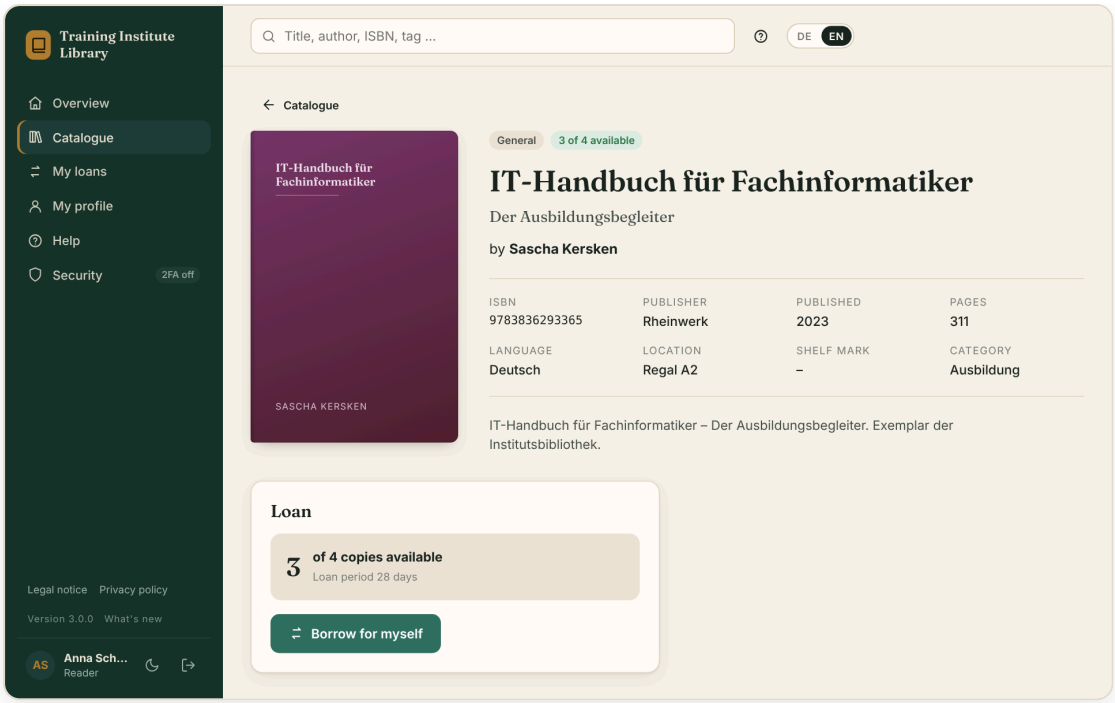


Fig. 9 – Book page with details, location and the button for borrowing.

IF THE BUTTON IS MISSING

- All copies are currently on loan – then you can **reserve** it (next section).
- The book belongs to a group you are not a member of.
- The book is **reference only**: it is not lent out, only used on site.
- You have already borrowed the book or reached the maximum number of loans.
- Self-checkout is turned off – then the library lends books at the circulation desk.

Reservations

If all copies are on loan, the book page shows **Reserve** (if the library has switched reservations on). You join a queue and see your position.

1. When a copy is returned, it is **set aside** for the first person in the queue. You get an email, and the **Overview** says it is ready for you.
2. Collect the book within the collection period (default 7 days) – borrow it yourself or at the desk. After that it is the next person's turn.
3. Your reservations and positions are shown under **My loans**, where you can also delete them.

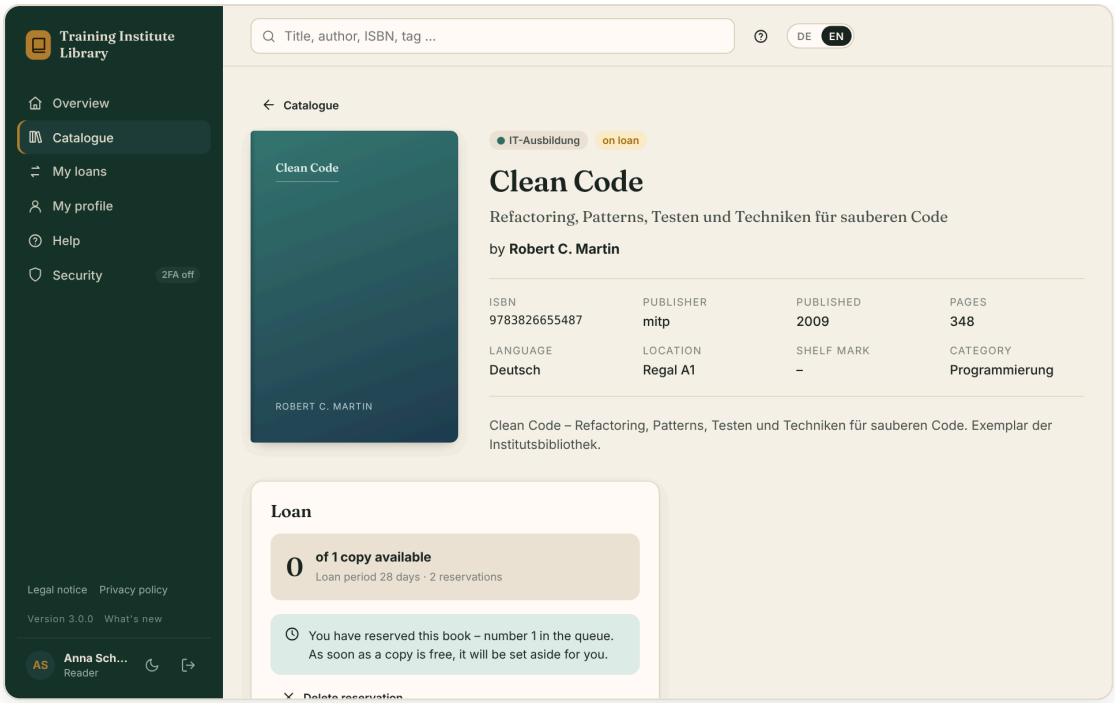


Fig. 10 – A reserved book: position 1 in the queue.

> **GOOD TO KNOW**
> You cannot renew a reserved book yourself – others are waiting for it. If a reservation no longer applies (account locked, no longer in the book's group, book reference only), it is closed automatically and the copy goes to the next person. Nobody can borrow a freed copy past the queue.

My loans

Under **My loans** you see everything you currently have, with the loan date and due date. The tabs at the top show only overdue loans or your past history.

- **Renew** extends the due date by 14 days – at most twice per loan (default values that the library can change). It counts from the later of the two dates: today or the current due date.
- **Return** only appears if the library allows self-return. Otherwise, bring the book to the circulation desk.

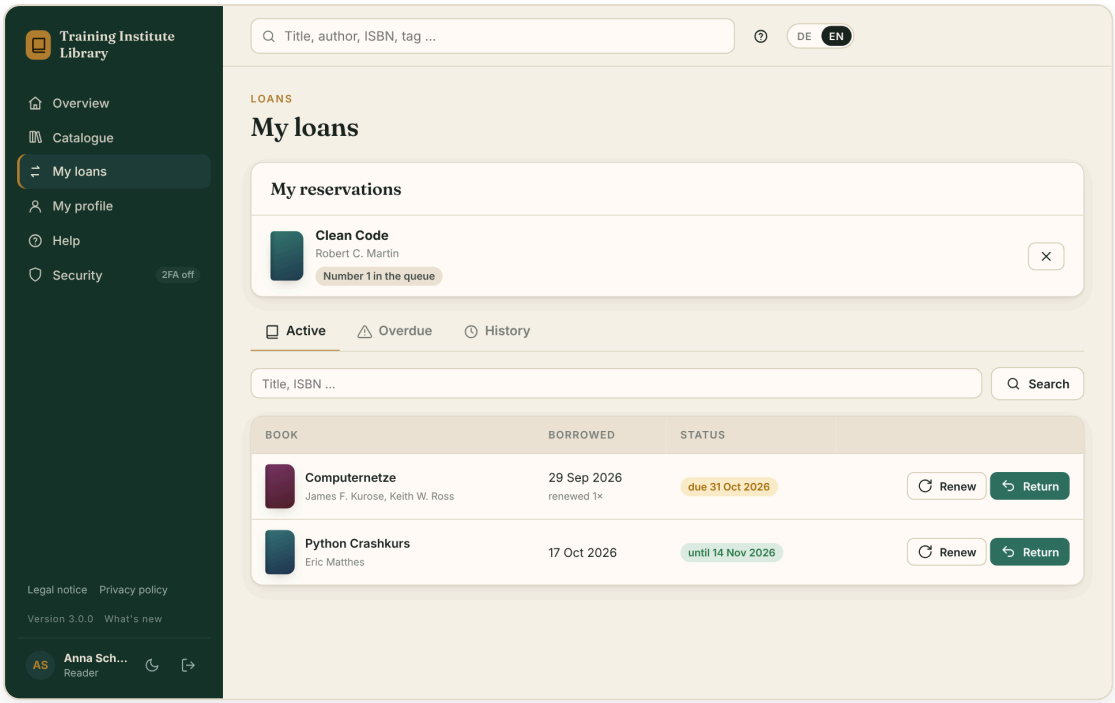


Fig. 11 – Your own loans with due date, Renew and Return.

Profile and emails

Under **My profile** you find your loan history and account details. This is also where you turn **email notifications** on or off: confirmations, a reminder shortly before the due date, and overdue notices. Local and self-registered accounts change their display name and email address themselves under **My details** (after self-registration a new address only applies once confirmed by link) and set a new password (at least 10 characters) under **Change password**. For network accounts (Active Directory), name, email and password come from the institute's directory and are maintained there.

> **LOST YOUR PHONE?**
> Under *Profile → Security → Signed-in devices*, **Sign out on all other devices** signs you out everywhere else. This also happens automatically when you change your password.

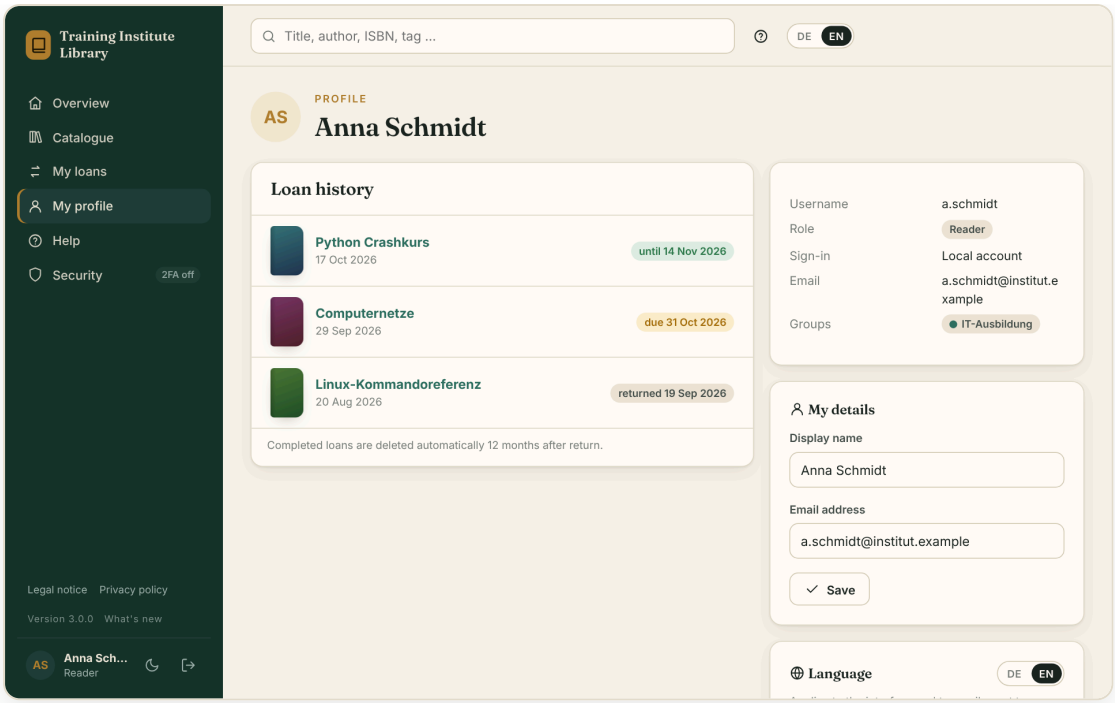


Fig. 12 – The profile of a local account with loan history, account details and "My details".

> **PRIVACY**
> Completed loans are deleted automatically after a fixed period (default: 12 months after the return). The note below your history states the current period.

Setting up two-factor authentication

A second factor protects your account even if someone knows your password. It may be mandatory for librarians and administrators – in that case the program takes you here automatically after signing in. You will find this area under **Security** in the sidebar.

Authenticator app (recommended for everyone)

1. Install an authenticator app on your phone, e.g. Microsoft Authenticator, Google Authenticator, Aegis or 2FAS.
2. Under *Security*, click **Set up app**.
3. Scan the QR code with the app. If that doesn't work, type in the key shown.
4. Enter the 6-digit code from the app and click **Confirm**.
5. **Important:** write down or print the **10 recovery codes** shown and keep them in a safe place. Each code works exactly once.

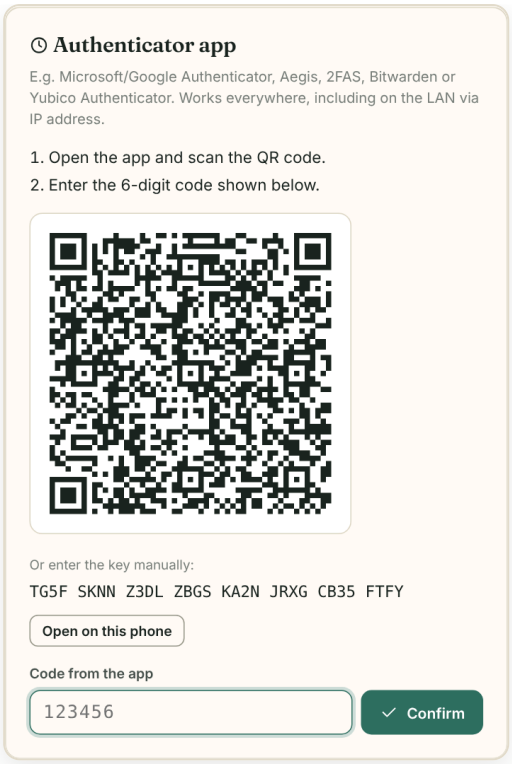


Fig. 13 – Setting up the authenticator app with a QR code.

Security key (YubiKey, passkey)

Give the key a name (e.g. “blue YubiKey”), click **Add** and touch the key when it flashes. Ideally, register a **spare key** straight away. Security keys only work via the library's encrypted address (https) – and only at the address where they were registered.

> **RUNNING OUT OF CODES**
> If your recovery codes are running low, generate new ones under *Security*. The old ones then stop working.

Registering yourself

If the library allows it, the sign-in page shows a **Register now** link. New accounts are always reader accounts. Depending on the settings, you have to confirm your email address via a link (valid for 48 hours) and/or wait for the library to approve your account – you will then receive an email. The password must be at least 10 characters long.

CHAPTER 4

For librarians: collection and lending

Librarians look after the collection and lend books. The additional menu items are in the sidebar under *Management*.

Adding a book

1. Click **Add book** (or **+ Book** at the top).
2. Enter the **ISBN** – by keyboard, with a **USB or Bluetooth barcode scanner** (just click into the field and scan) or with the **camera** via the **Camera** button.
3. Click **Fetch data**. Title, author, publisher, year, number of pages and cover are retrieved automatically from the German National Library (DNB), Open Library and Google Books. Please check them briefly.
4. On the right under **Holdings**, choose the **group** (or *General* for everyone), the number of **copies**, the **shelf mark** and the **location**.
5. **Save** – or **Save & next** to add the next book straight away.

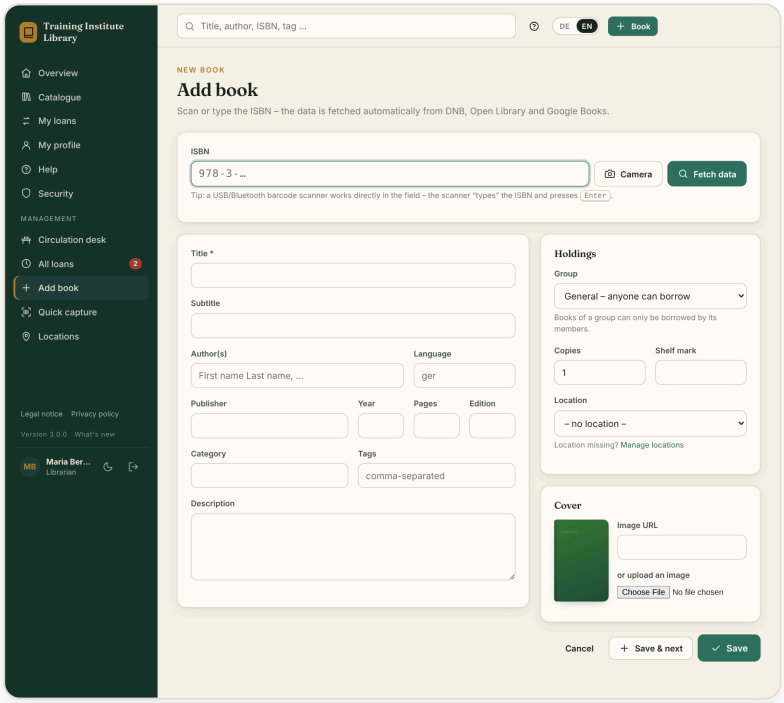


Fig. 14 – Add book: enter the ISBN, fetch data, choose group and location.

TIPS

- If the ISBN is already in the collection, the program points this out. If you still save the book in the **same group**, no second entry is created: the copies are added to the existing book (the other details in the form are not applied). In a different group a separate entry is created.
- The camera only works via the encrypted address (https). A USB barcode scanner always works and is the fastest option at the desk.
- You can upload your own cover photo in the *Cover* section.

Quick capture

For many books at once – for example when setting up the library or when a delivery arrives. On the right you set the **group** and **location** for all the following books. Then you scan one ISBN after another:

- **New title:** the program shows the data it found, and Enter adds it. With **Add new titles automatically (without asking)** it works without any prompt at all.
- **Already there:** a click on **+1 copy** increases the number.
- **Nothing found:** **Add manually** opens the normal form with the ISBN.

The log on the right lists everything done in the session, with links to the books.

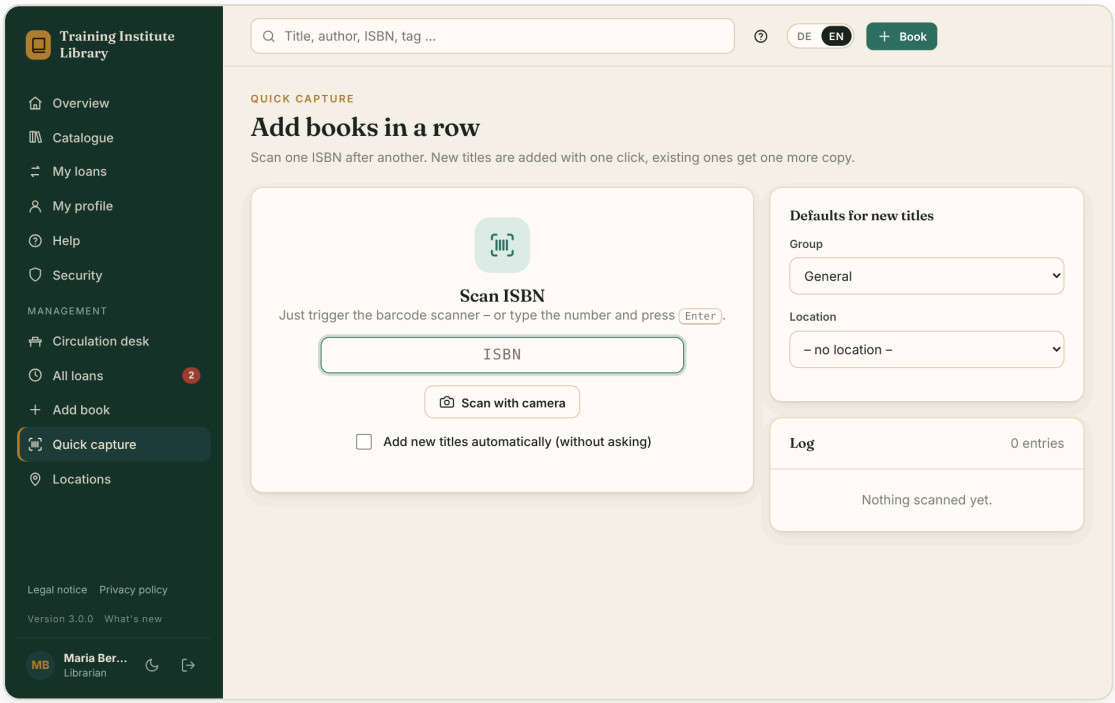


Fig. 15 – Quick capture: scan, Enter, next book.

Locations

Under **Locations** you define your shelves, rooms and cabinets once – several at a time, one per line. As soon as there is at least one location, it is **picked from a list** instead of being typed when adding books, in quick capture and when editing. So there are no more typos.

- **Rename:** change the name in its field and save with ✓ – all books at that location are updated too.
- **Deleting** is only possible once no book is left there.
- **In the holdings, but not defined:** this lists locations typed freely in earlier entries. Add them as a location or replace them with a defined one. For a different spelling (“shelf a2”) or a typo (“Shlef A1”), the matching location is already preselected – one click on **Apply** is enough.
- The number next to each location leads straight to the books there.

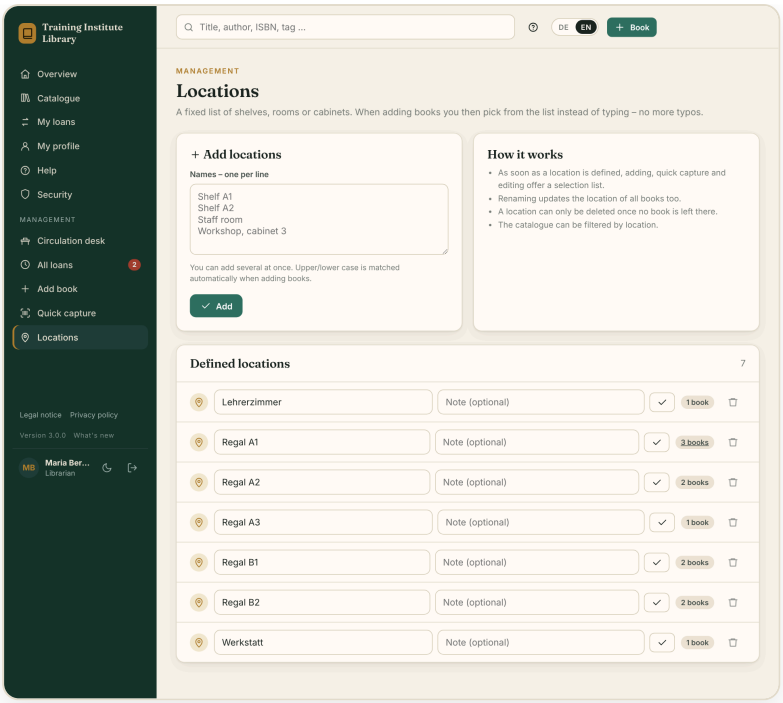


Fig. 16 – Adding and managing locations.

Editing and deleting books

On the book page you will find **Edit** and **Delete**. A book that is currently on loan cannot be deleted. The number of copies cannot drop below the number on loan. At the bottom of the book page is the **history** of all loans of this book; older loans without individual details are only counted.

With **CSV** at the top of the catalogue you can download the whole collection as a spreadsheet (opens in Excel).

Lending from the catalogue

Under every available book you manage, the catalogue shows a **Lend** button (in the list view at the right of the row). It opens the book page right at the **Lend to a user** form with the cursor already in the **User** field: type a name, username or email, pick the person from the list, optionally a due date and a note, then **Lend**.

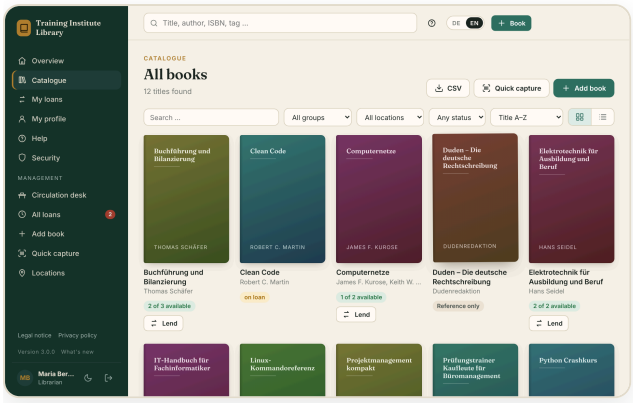


Fig. 17 – The catalogue as seen by a librarian: "Lend" under every available book.

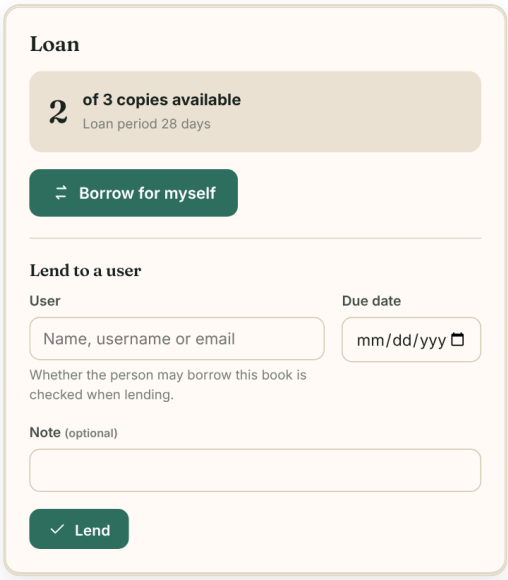


Fig. 18 – The form on the book page.

The circulation desk

The **Circulation desk** is made for the counter: the input field stays active, so you simply scan one item after another.

Loan

1. **Select the user:** type a name, username or email – matching people appear in a list right away (pick with the arrow keys and `Enter` or by clicking). With Active Directory connected, the search also finds people who have never signed in; their account is created when you first lend them a book. On the right you see what the person already has on loan and what is set aside for them.
2. **Scan or search for the book:** scan the ISBN or shelf mark – or type a title, author or tag: matching books of your collection appear with location and available copies. Enter the internal book number with a leading `#` (e.g. `#12`) – plain digits are treated as a shelf mark.
3. With **Lend immediately on an exact match (ISBN, shelf mark)**, every scanned book is lent without another click – ideal with a barcode scanner. A typed title never lends automatically.

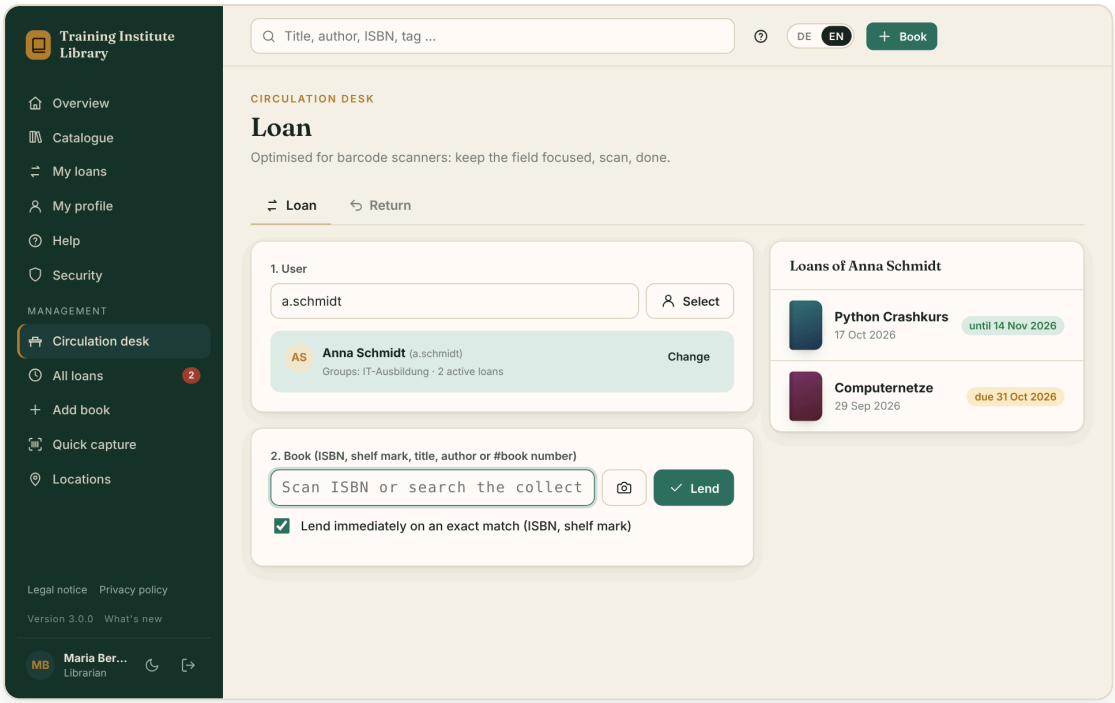


Fig. 19 – Circulation desk: person selected, scan the book.

The program automatically checks whether the person may borrow the book (group, maximum number of loans, account active), whether a copy is available and not set aside for a reservation, and tells you the reason if not. Books that are **reference only** are never offered for lending at the desk.

Return

In the Return tab, all you need to do is scan the returned book. If exactly one copy is on loan, an exact match (ISBN, shelf mark) checks it in straight away; otherwise you choose the matching loan from the list. If the book is reserved, the desk says "Please set aside – reserved for …" – then put the book aside instead of back on the shelf.

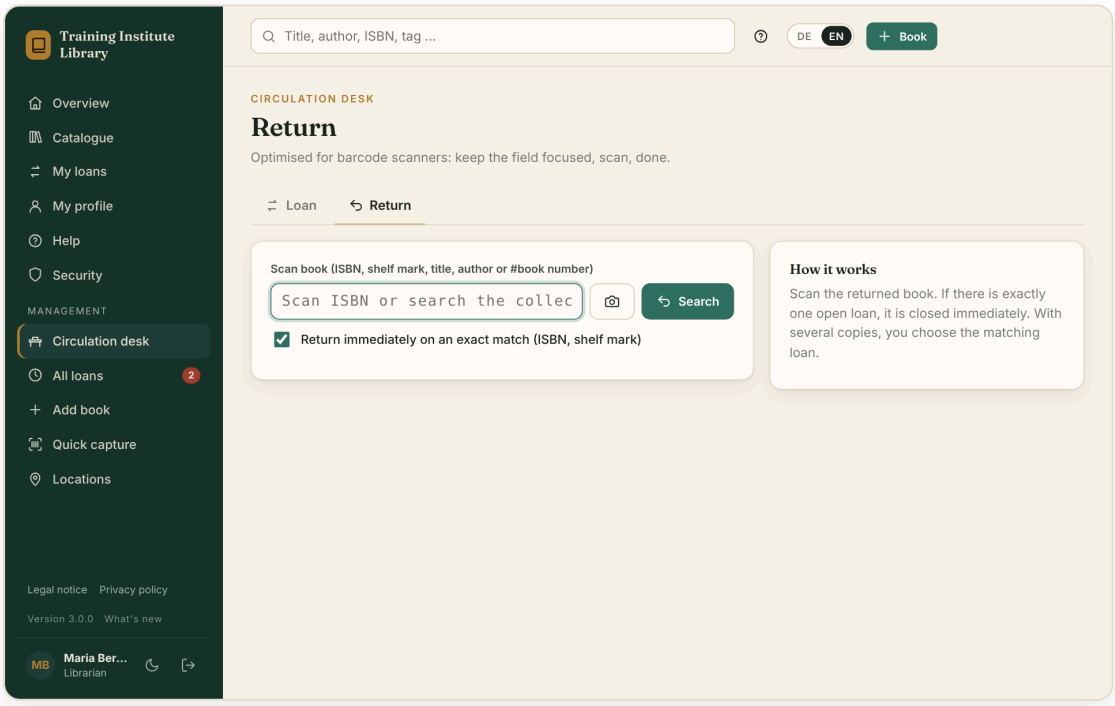


Fig. 20 – Returning a book at the desk.

Managing reservations

The book page shows the **queue** with names, dates and status (*waiting* or *set aside until …*), and you can delete single reservations. When a person with a book set aside comes to the desk, it appears at the top under "Set aside for …" after selecting them – one click lends it. An administrator switches reservations on under *Settings → Reservations*, where the collection period and the maximum per person are set too.

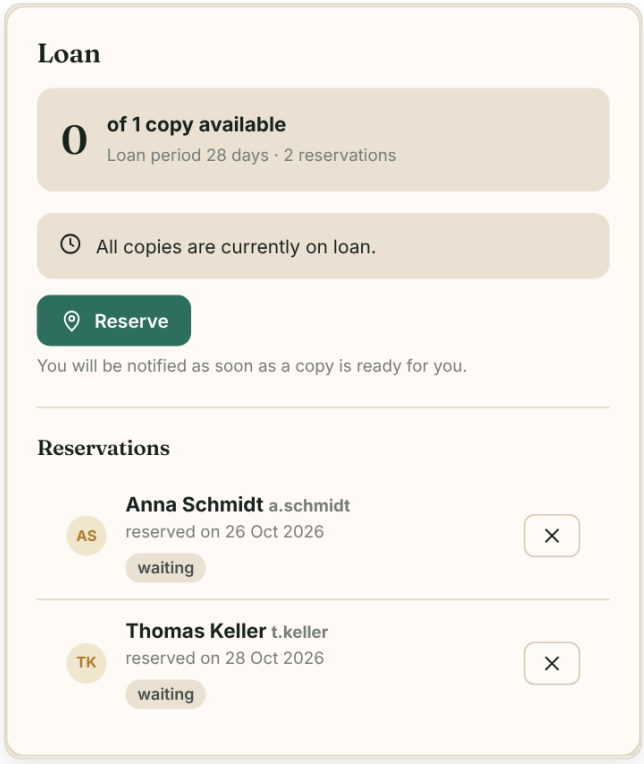


Fig. 21 – The queue on the book page.

All loans and overdue notices

Under **All loans** you see all current loans in your area, with tabs for *Overdue* and *History*. The red number in the sidebar counts the overdue loans.

- **Renew** – librarians may also renew beyond the maximum number of renewals.
- **Envelope** – sends a reminder by email right away.
- **Return** – checks the book back in.

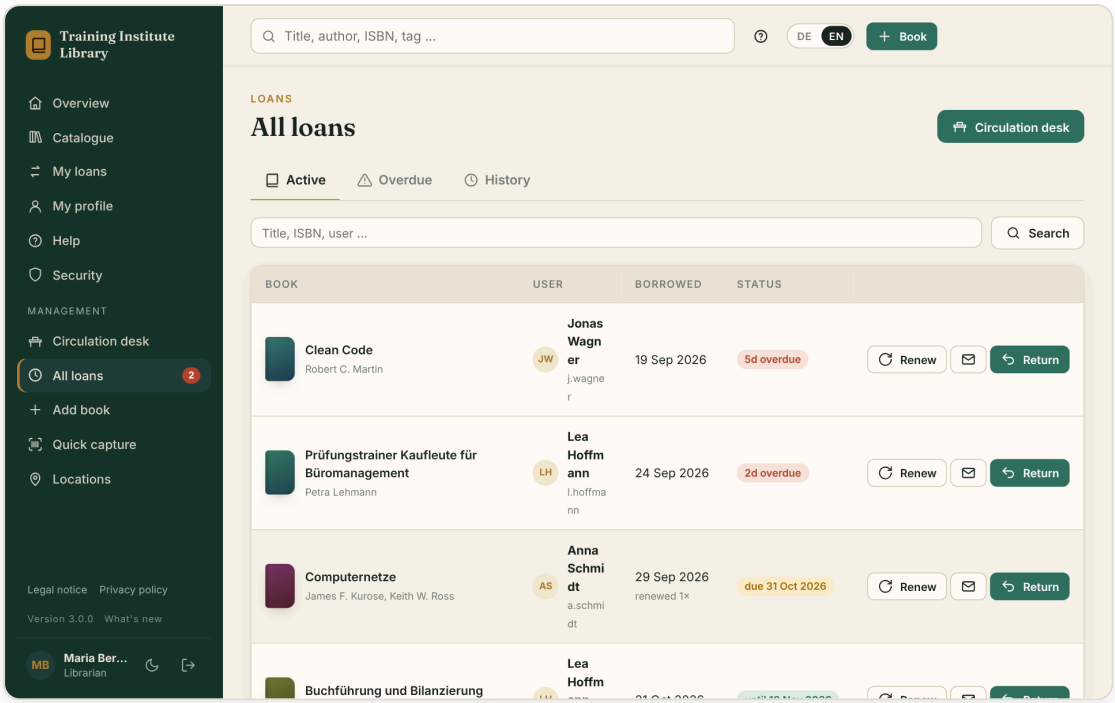


Fig. 22 – All loans with overdue entries (red).

The program also sends reminders before the due date and overdue notices automatically once a day if email is set up (see section 5.5).

Group librarians

If a librarian is assigned to one or more groups, they only look after the books of **those groups** and the general books, and only see their loans. A librarian **without** a group is responsible for the whole collection. All librarians may add locations; renaming that would affect books of other groups is reserved for administrators and librarians without a group.

CHAPTER 5

For administrators

Administrators set up the library and take care of accounts, security and operations. All areas are in the sidebar under *Administration*.

Users

The user list shows all local accounts and all network accounts that have signed in at least once – with role, source (*local* or *AD*), groups, current loans and last sign-in. Labels point out special cases: *awaiting approval*, *locked*, *Sign-in lock*, *2FA*. You can filter by these using the selection box.

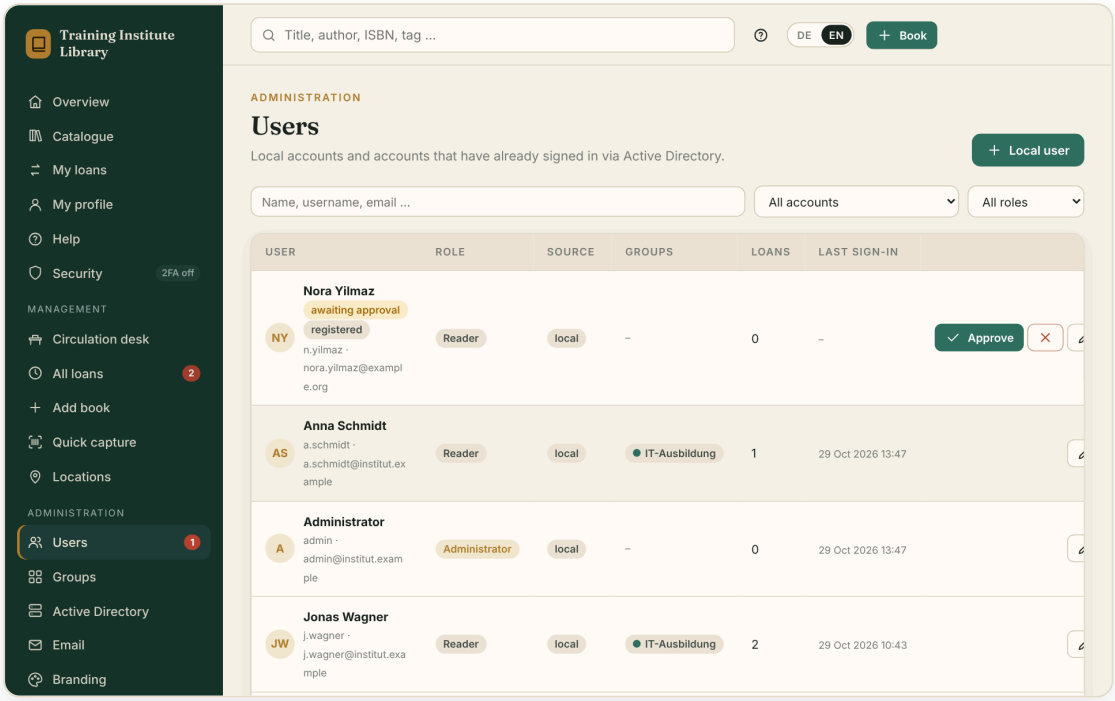


Fig. 23 – User list with a registration awaiting approval.

- **Local user** creates an account without network sign-in – for guests, say, or as an emergency admin.
- **Approve** / **Reject** appear for self-registrations awaiting approval.
- **Edit** opens the account: name, email, **role**, **groups**, new password (local accounts only), notifications and **Account active**. A deactivated account cannot sign in or borrow anything.

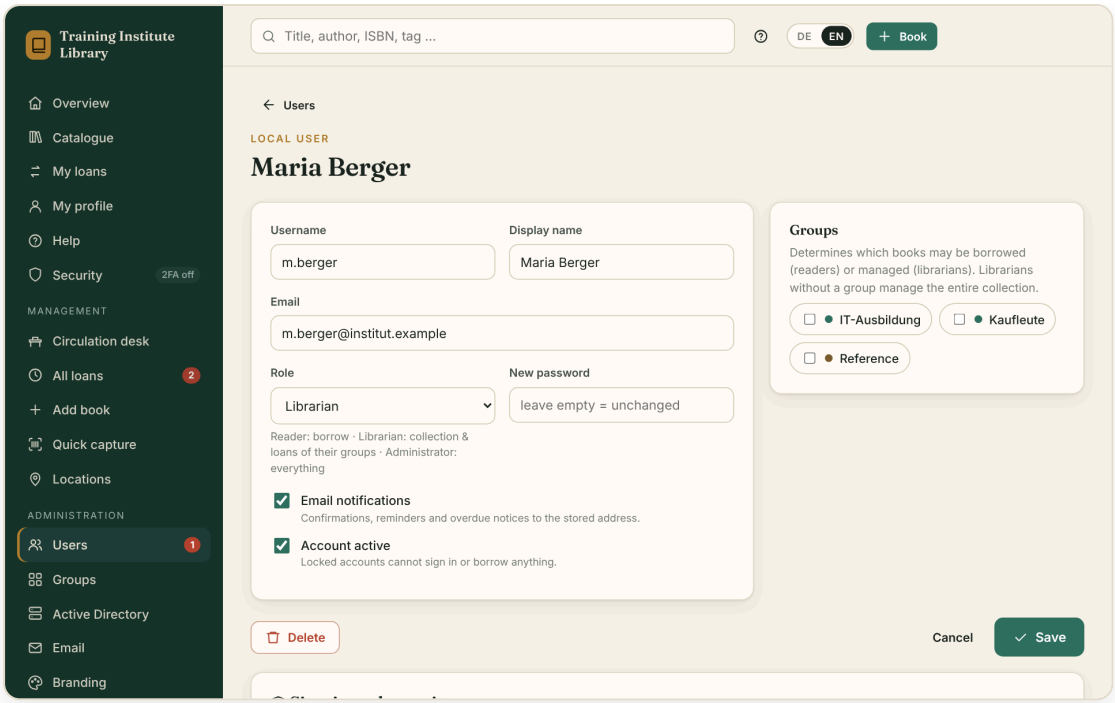


Fig. 24 – Editing an account.

> **NOTE**
> For network accounts, role, name, email and (if set up) groups are taken from Active Directory at every sign-in. So it's better to make changes to them in AD.

Solving sign-in problems – without the command line

At the bottom of the account is the **Sign-in and security** section:

- **Reset** next to two-factor authentication removes the authenticator app and security keys. The person then signs in with just their password again; if a second factor is mandatory, they are guided through setting it up again.
- **Lift lock** unlocks an account immediately after too many failed attempts (otherwise this happens by itself after 5 minutes).

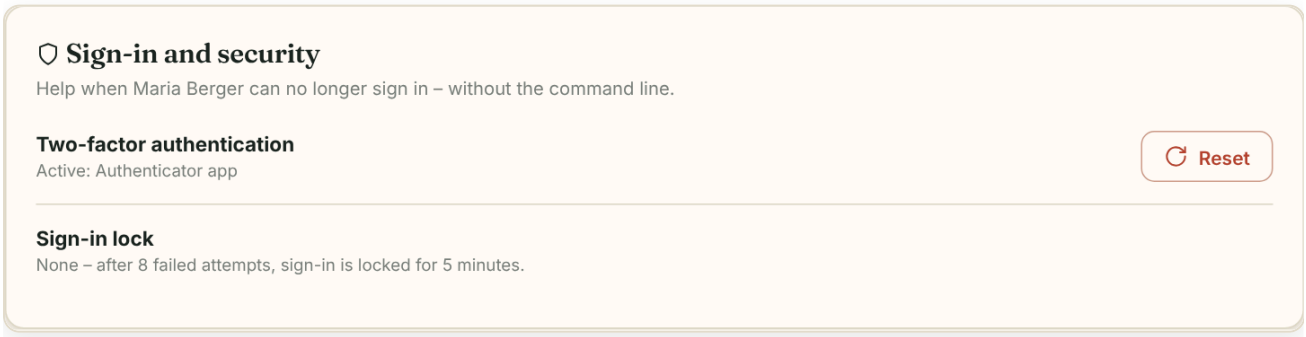


Fig. 25 – The “Sign-in and security” section in the user account.

ALL ADMINS LOCKED OUT?

If “no two-factor prompt” is set for the internal network, an admin can get in from the internal network without a second factor. Otherwise IT can help with a command on the server (section 6.6). Safest of all: have **two** administrators and keep the recovery codes in a safe place.

Groups and reference-only stock

Under **Groups** you create groups, e.g. one per apprenticeship programme. Each group has its own collection; members only borrow books of their groups and general books. You add members in the group – or you link the group to an **AD group**, and members are then added and removed automatically at every sign-in.

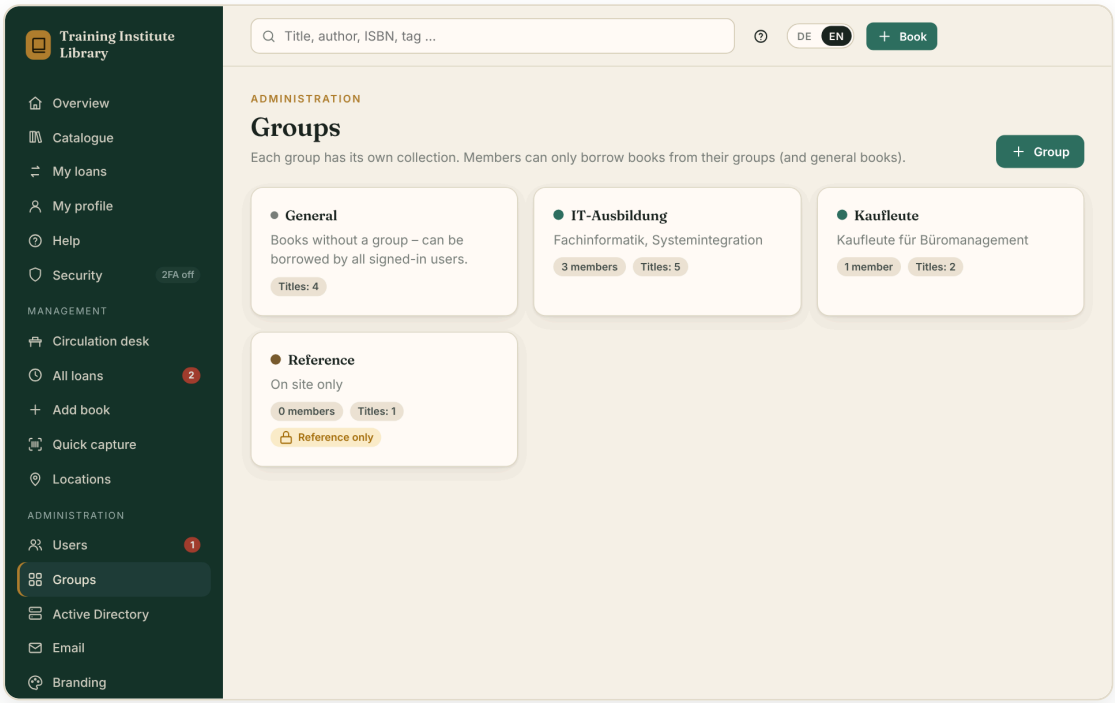


Fig. 26 – Groups with members and titles.

Reference only: with **Reference only – books in this group are not lent out** ticked, the group's books stay visible in the catalogue (marked) but are no longer offered for lending anywhere – not at the desk, not on the book page, not for self-checkout and not for reservations. Open reservations for these books are closed when you save; copies already on loan are returned as usual. When saving, the group takes exactly the ticked members.

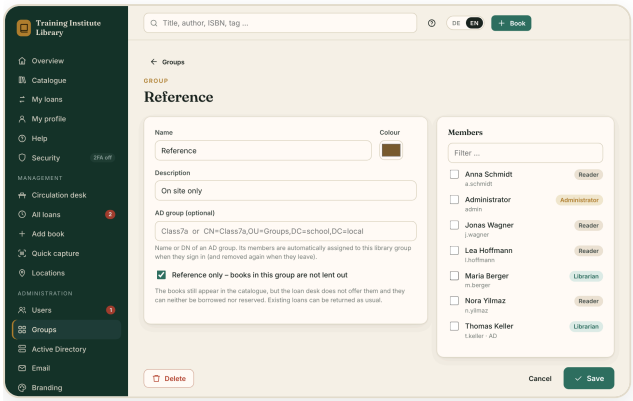


Fig. 27 – A reference-only group.

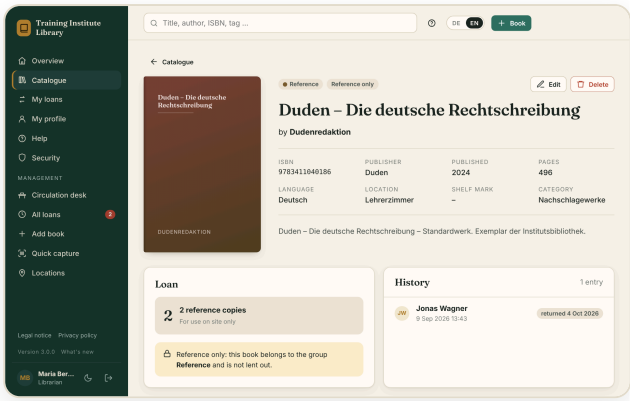


Fig. 28 – Book page of a reference-only book.

Active Directory

With the institute's Active Directory (Windows Server or Samba), everyone signs in with their usual credentials. Setup is usually done by IT:

1. **Connection:** enter the server (e.g. `ldaps://dc1.institute.local`), security, base DN and a service account with read permissions.
2. **Certificate:** with LDAPS/StartTLS, **Fetch certificate from DC** retrieves the certificate automatically; check the fingerprint and click **Trust & save**.
3. **Roles:** under *Who can do what?*, enter the names of the AD groups for administrators, librarians and readers. **Load groups** lists the AD groups for you to click. Nested groups count too; anyone who is in none of the groups is rejected.
4. **Test:** **Test sign-in** shows all groups of an account and the role it would get – without saving first.

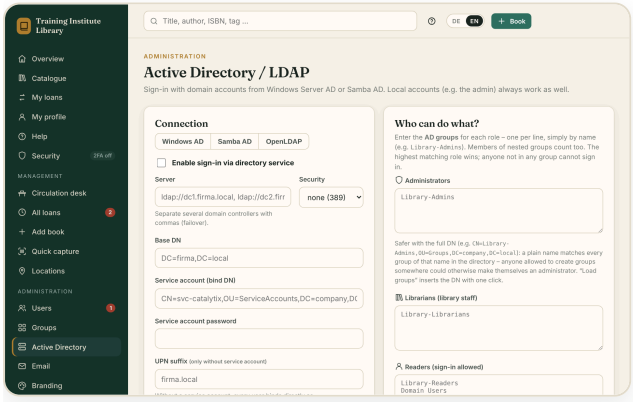


Fig. 29 – Connection to Active Directory.

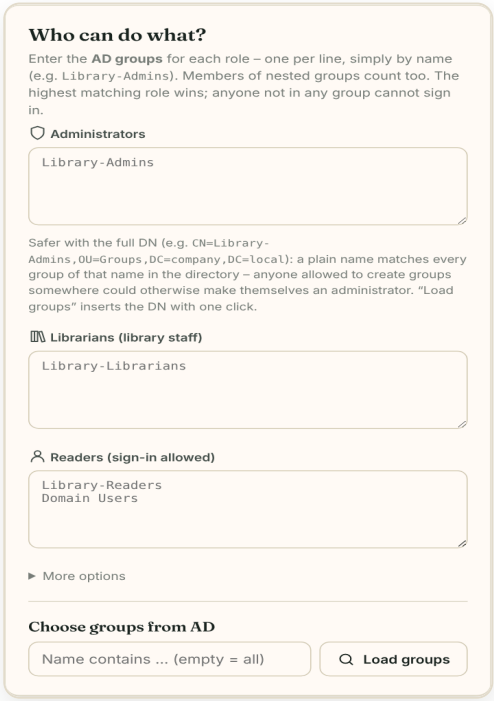


Fig. 30 – Roles via AD groups.

Directory sync and people cache: with a service account the program fetches everyone allowed to use the library in the background (hourly by default). The people search at the desk then finds them instantly. Accounts disabled, deleted or no longer entitled in AD are locked (not deleted) and unlocked automatically when the person comes back; renames in AD are recognised. A safety brake stops a faulty sync from locking many accounts at once.

> **NOTE**
> Local accounts always take precedence. That's why the local administrator account still works when Active Directory cannot be reached.

Email

Under **Email** you enter the mail server (SMTP) and choose which messages are sent. **Send test email** checks the settings.

MESSAGE	WHEN
Confirmation on loan	immediately on loan, with title and due date
Confirmation on return	immediately on return (off by default)
Reminder before due date	a few days before the due date (default: 2), once per loan
Overdue notice	from the first overdue day, repeated every 7 days, at most 3 times (adjustable)
Daily overview for librarians	daily, only if there are overdue loans in their groups (off by default)
Backup failed	to all administrators

Reminders and overdue notices go out once a day from the time you set; several books of one person end up in **one** email. The **Library address** is important so that links in the emails work. At the bottom of the page is the **Delivery log** – it is cleared automatically after the retention period.

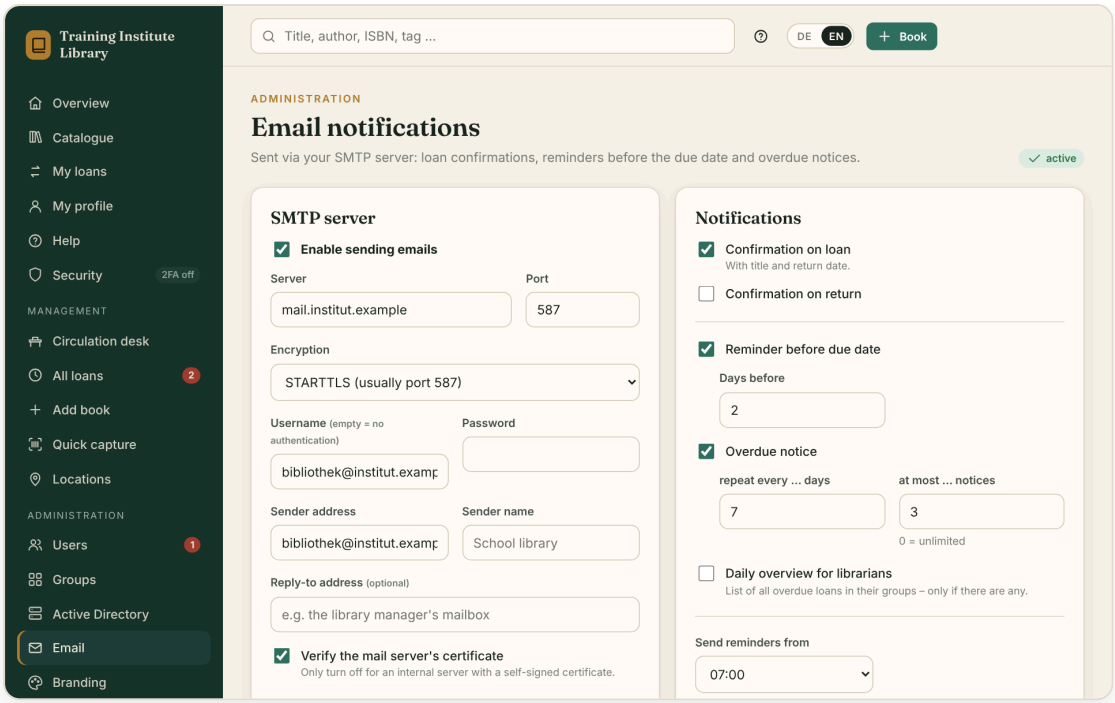


Fig. 31 – Email settings.

Branding

Under **Branding** you adapt the library to your organisation – with a live preview:

- **Logo** (PNG with a transparent background, or SVG) – appears in the sidebar, on the sign-in page, in emails and as the browser icon.
- **Background image** of the sign-in page, with adjustable darkening.
- **Colours** for buttons, accents and sidebar – text colours are chosen automatically so that everything stays readable.
- **Texts** on the sign-in page. You set the **library name** under *Settings*.

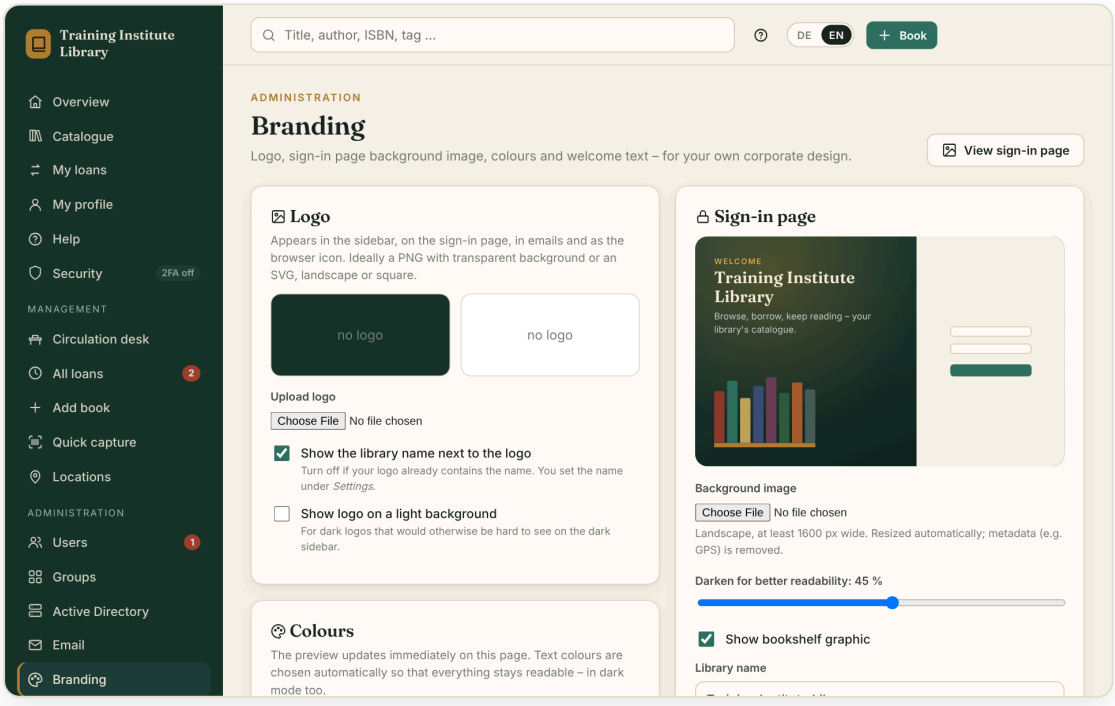


Fig. 32 – Branding with a preview of the sign-in page.

> **NOTE**
> The logo and background image are publicly accessible because they are already needed on the sign-in page – so don't use anything confidential.

Settings

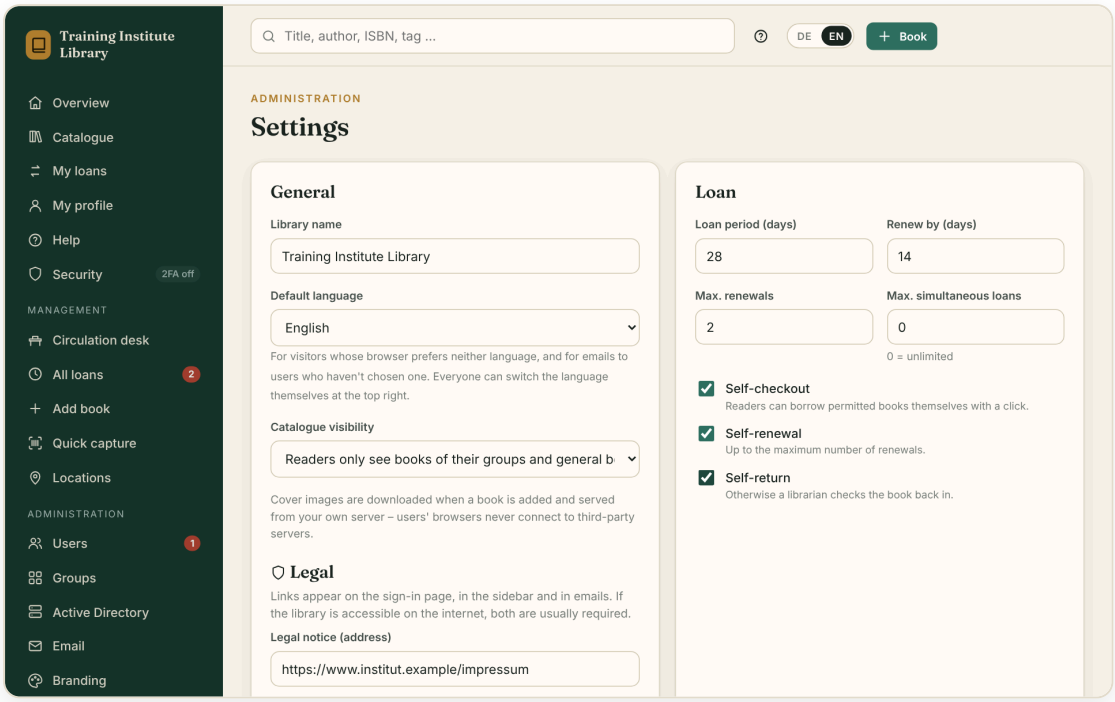


Fig. 33 – General settings and loan rules.

SECTION	WHAT YOU SET
General	Library name; whether readers see the whole catalogue or only their groups and general books.
Legal	Addresses of the **legal notice** and **privacy policy** (e.g. on the institute's website). They appear on the sign-in page, in the sidebar and in every email. Usually mandatory if the library is accessible on the internet.
Loan	Loan period (default 28 days), renewal (14 days, at most twice), maximum number of simultaneous loans (0 = unlimited), self-checkout, self-renewal, self-return.
Reservations	on/off (default off), collection period (default 7 days), maximum open reservations per person (default 5). Switching off closes all open reservations.
Self-registration	off (default) / with approval by administrators / account active immediately; email confirmation, allowed email domains, group for new readers.
Retention	Deletion period for completed loans and the email log (see 5.9).
Two-factor authentication	Mandatory from outside, rule on the internal network (see 5.8).

ISBN data sources	Which catalogues are queried, and in which order.

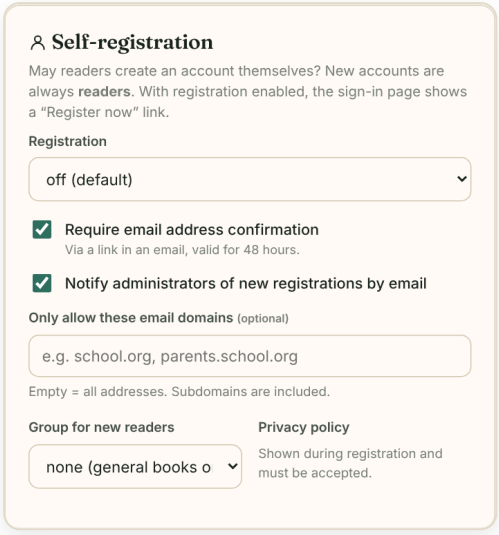


Fig. 34 – Self-registration.

Two-factor and internal network

The two-factor rules can be set separately for access from the internet and from the internal network:

- **Mandatory for access from outside:** nobody, administrators, librarians and administrators, or everyone.
- **On the internal network:** *same as from outside* (default), *not mandatory* (users who have 2FA are still asked) or *no two-factor prompt*.
- **Internal network:** your networks, one entry per line, e.g. `192.168.2.0/24`. Empty means 10.x and 192.168.x; Docker networks (172.16–31.x) only count if you enter them. Better to restrict it to your actual LAN.
- **Own internet connection:** static IP or DynDNS name. Needed when internal computers open the library's public address and the router passes them on with its own public IP. DynDNS names are re-resolved every 5 minutes.

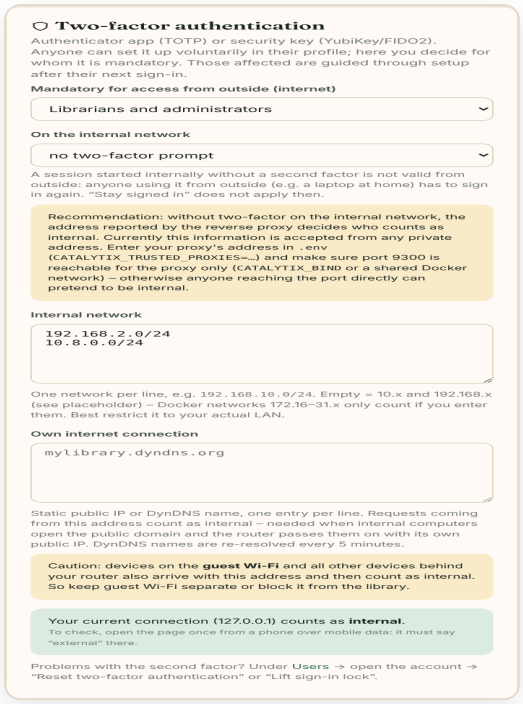


Fig. 35 – Two-factor rules and internal network.

SET UP CORRECTLY?

At the bottom of the section it says whether **your current connection** counts as internal or external. Open the page once from a phone over mobile data (Wi-Fi off): it must say **"external"** there. From a computer on site it should say **"internal"**.

GUEST WI-FI

If your own internet connection is entered, devices on the guest Wi-Fi also count as internal, because they go online via the same public address. Guests would then only need a password. A cleaner solution is an internal DNS entry that points the library address to the server on the LAN when on site.

SECURITY

A session started on site **without** a second factor is not valid from outside. "Stay signed in" has no effect in this case.

Retention and data protection

Under the General Data Protection Regulation (GDPR), personal data may only be stored for as long as necessary. That's why the program automatically deletes:

- **completed loans** the set time after the return (3, 6, **12**, 24 or 36 months). Only the number of loans per book is kept.
- **entries in the email log** after the same period – except those relating to current loans, because that is how the program knows how many overdue notices have already been sent.
- **unconfirmed self-registrations** after 7 days.

Current and overdue loans are never deleted. The clean-up runs once a day; **Clean up now** starts it immediately.

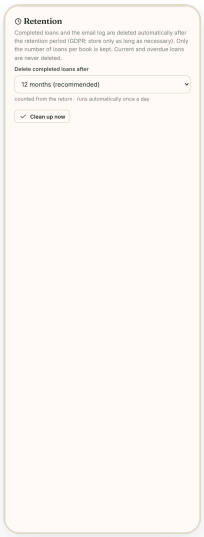


Fig. 36 – Retention period.

FOR YOUR PRIVACY POLICY

The following is stored: name, username, email address, group membership, loans with dates, the email log and – with two-factor authentication – the registered keys. The retention period and how long backups are kept (with 10 daily backups, up to 10 days longer) should also be included. The program does not load any content from third-party servers: fonts, covers and barcode detection come from your own server, there is no tracking and only technically necessary cookies are used.

Backup

The database is **backed up automatically every day** (by default from 2 a.m.), while the program is running, and every backup is checked for damage. The **10 most recent backups** are kept; the number can be set from 1 to 100, and older ones are deleted automatically.

- **Back up now** creates a backup immediately – useful before major changes or an update.
- Every backup can be downloaded or deleted.
- If a backup fails, a **!** appears next to the menu item and all administrators receive an email.
- **Before every version change** the program backs up at start-up itself ("before update to X.Y.Z"). The last three of these are kept in addition to the daily ones – they are the way back to an earlier version (section 6.3).

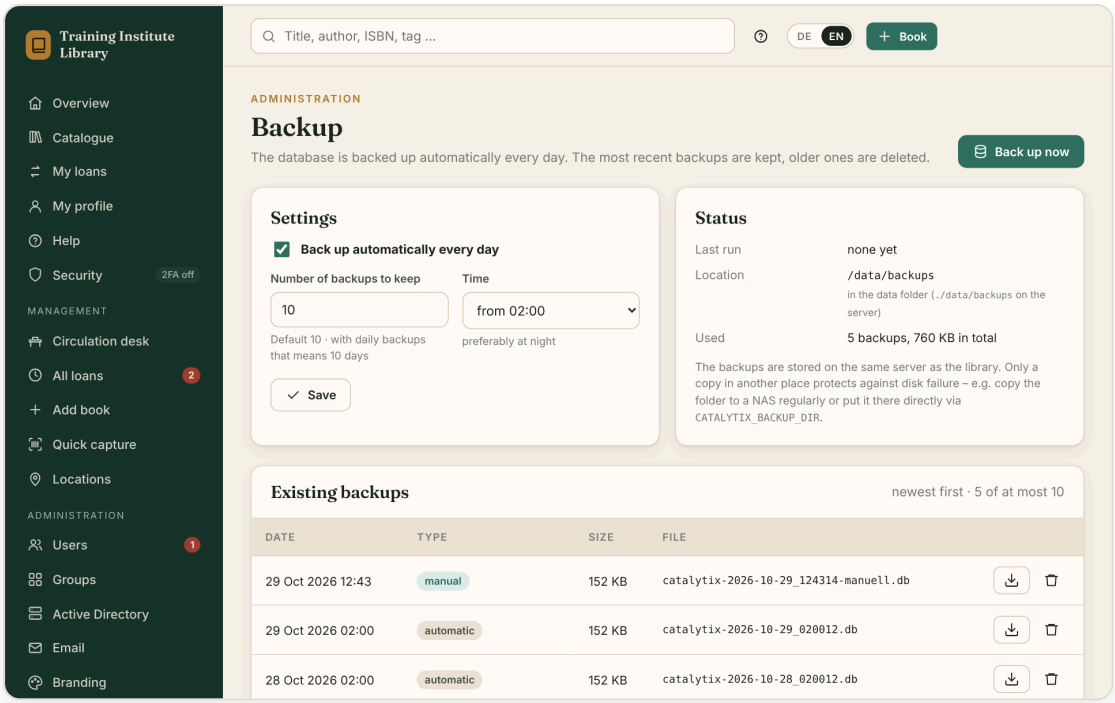


Fig. 37 – Backup with settings, status and existing backups.

IMPORTANT

The backups are stored on the same server. They protect against operating errors and failed updates, but not against a disk failure. Have the backup folder copied to another place regularly, e.g. to a NAS.

CHAPTER 6

Operations for IT

A short guide to setting up the server. All details are in the README in the repository (github.com/soleil0815/catalytix).

Installation

The program is completely inside the ready-made image `ghcr.io/soleil0815/catalytix` (Intel/AMD and ARM, e.g. Synology or Raspberry Pi) and listens on port **9300**. The server only holds `docker-compose.yml`, `.env` and the data folder `./data` – no program folder, nothing to unpack or build.

```
mkdir -p /opt/catalytix && cd /opt/catalytix
curl -fsSLO https://github.com/soleil0815/catalytix/releases/latest/download/docker-compose.yml
curl -fsSL  https://github.com/soleil0815/catalytix/releases/latest/download/env.example -o .env
chmod 600 .env
docker compose up -d
```

- **Your own values** (port, time zone, database …) go into `.env` – everything in it is optional. Additions such as a NAS share for backups go into your own `docker-compose.override.yml`. Updates never touch either.
- **First visit:** open `http://SERVER:9300` (e.g. `http://library.company.local:9300`) and create the local administrator account. The page asks for the **setup code**: `docker compose exec catalytix docker-entrypoint.sh flask --app wsgi setup-code`.
- **HTTPS:** for use over the internet, the camera and security keys, put the program behind a reverse proxy with a certificate (e.g. Nginx Proxy Manager, Caddy). The proxy must send `X-Forwarded-For` – the ones mentioned do this by default. Do **not** open port 9300 on the router.
- **Built safely:** base image, operating system and Python packages are pinned to exact versions with checksums. Every image is tested and scanned for vulnerabilities before it is published; every Monday a security update appears automatically when needed (`X.Y.Z-s1`, `-s2` …).

Updating

With a container manager (Dockhand, Watchtower, Tugtainer …) the library is updated like any other container. Recommendation: schedule updates **at night** – the library is unavailable for a moment while a new version starts. **By hand:**

```
docker compose pull && docker compose up -d
```

Or with `update.sh` (in the release; put it into the same folder). It does the same and in addition keeps the running image as `previous` and waits until the new version reports healthy. **If it does not start**, it restores the backup from before the update and starts the previous version again (asks first, automatic with `-y`). If the image brings a newer `docker-compose.yml`, it puts it in place and keeps the old one as `docker-compose.yml.alt`.

```
./update.sh                    # newest version (or the one fixed in .env)
./update.sh --version=3.0.0    # exactly this version (also a security update 3.0.0-s1)
./update.sh --load catalytix-3.0.0-amd64.tar.gz   # without internet: image file from the
release
```

WHAT THE PROGRAM DOES ITSELF

Before a new version changes the database, it backs it up ("before update to X.Y.Z"; with MySQL/PostgreSQL please back up yourself). If an **older** version starts on a database a newer one has already used, it stops with a clear message instead of running into follow-up errors. If the `docker-compose.yml` on the server is too old, the log says so and administrators see it on the overview.

Without internet: the release contains the images as files (`catalytix-X.Y.Z-amd64.tar.gz` , `-arm64` for ARM) with `SHA256SUMS` . `update.sh --load` checks the checksum, loads the image and starts it.

Fixing a version and going back

Fixing: set `CATALYTIX_TAG=3.0.0` in `.env` (or `./update.sh --version=3.0.0`) – it stays on this version, also with a container manager. `CATALYTIX_TAG=latest` follows every new version again.

Going back: an older version needs the database from back then. `./update.sh --version=X.Y.Z` restores the newest backup this version can read and starts it – it asks first, because everything entered since is lost. By hand:

```
docker compose stop catalytix
docker compose run --rm -e CATALYTIX_SKIP_DB_CHECK=1 catalytix flask --app wsgi restore-for-
version 3.0.0
# in .env: CATALYTIX_TAG=3.0.0
docker compose up -d
```

Moving from versions before 3.0

Up to 2.20 the program files were on the server (zip package, built locally). Once, in the existing program folder:

1. Take a backup: *Administration → Backup → Back up now*.
2. Switch to the image with the **existing** script: `./update.sh --pull` . It gets 3.0, places the new `docker-compose.yml` , `.env.example` and the new `update.sh` , and starts it.
3. Remove the old program files: `./update.sh --cleanup` . It shows the list (only known program files such as `catalytix/` , `Dockerfile` , old zip packages and locally built images), asks, and deletes only those. **Never** touched: `data/` , backups, `.env` , `docker-compose.yml` and anything a container mounts – not even through links.

Afterwards the folder only holds `docker-compose.yml` , `.env` , `.env.example` , `update.sh` and `data/` .

Restoring

To restore a backup, the library is stopped briefly. The current state is first backed up automatically as “before restore”, so a mistake can be undone:

```
docker compose stop catalytix
docker compose run --rm catalytix flask --app wsgi list-backups
docker compose run --rm catalytix flask --app wsgi restore-backup catalytix-2026-10-19_020012.db
docker compose start catalytix
```

To move completely to another server, take the whole `./data` folder with you – besides the database, it also contains covers, the logo, certificates and the key used to encrypt the two-factor data and the stored AD and mail passwords. The database and `secret_key` always belong together: if the key doesn't match the database, the program refuses to start and says so in the log.

Emergency commands

Run all commands in the container, prefixed with `docker compose exec catalytix docker-entrypoint.sh` – this runs them with the app's permissions instead of as root:

COMMAND	EFFECT
`flask --app wsgi reset-password admin`	set a new password for a local account
`flask --app wsgi reset-2fa admin`	reset an account's second factor
`flask --app wsgi create-admin notfall`	create an additional local administrator account
`flask --app wsgi backup`	back up the database immediately
`flask --app wsgi list-backups`	list backups (also "before update to …")
`flask --app wsgi send-reminders`	send reminders and overdue notices immediately
`flask --app wsgi apply-retention`	apply the retention period immediately

VERSION AND LICENCE

The *About this program* page (click the version number in the sidebar) shows the version, licence, the developer (helfi.space) and the link to the source code. The program is free software under the GNU AGPL v3.

CHAPTER 7

Frequently asked questions

Short answers to the questions that come up most often in everyday use.

QUESTION	ANSWER
Why can't I borrow a book?	All copies are on loan (then reserve it), it belongs to another group or is reference only, it is set aside for a reservation, you already have it or have reached the maximum – or self-checkout is turned off. The book page states the reason; if in doubt, ask at the circulation desk.
How often can I renew?	By default twice, by 14 days each time. After that, the library can renew once more.
I'm not receiving any emails.	Under *My profile*, check that an email address is stored and notifications are turned on. Also look in your spam folder.
The camera won't start.	The camera only works via the https address and needs your permission in the browser. At the desk, a USB barcode scanner is more reliable.
My YubiKey is not accepted.	Security keys only work for the address where they were registered. Via a different address (e.g. the IP), sign in with the authenticator app or a recovery code.
On site I'm still asked for the second factor.	Under *Settings → Two-factor authentication* the page shows which address the server sees. If it shows your public IP, enter your own internet connection or create an internal DNS entry (see 5.8).
A location is misspelled.	Rename it under *Locations* – all books there are updated too. Old typos are listed under “In the holdings, but not defined” and can be replaced with one click.
A book is in the collection twice.	Increase the number of copies on the original and delete the duplicate entry (only possible if it is not on loan). Since 2.20, newly added books with the same ISBN in the same group are added to the existing one automatically.
My reserved book is ready – what now?	Borrow it within the collection period (yourself or at the desk). The period is shown on the overview; after that it is the next person's turn.
The library does not start after an update.	With `update.sh` it goes back to the previous version automatically. Otherwise check the log (`docker compose logs catalytix`) and go back if needed (section 6.3).
Something was deleted by mistake.	Restore a backup from the previous day (section 6.5). Please note: everything that has happened since then will be lost.
Who can see my loans?	You, the librarians of your groups and the administrators. Completed loans are deleted after the retention period.